

Towards SAT-Based Learning of NNF Networks

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- ② SAT Encoding
- ③ Experimental Results
- ④ Conclusion

Motivating Example

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Example	cap-color red	bruises	foul odor	Poisenous
	Yes	No	Yes	Yes
	No	No	Yes	No
	No	Yes	No	Yes
	No	No	No	No

Goal: predict whether a new mushroom is poisenous or not.

Motivating Example – Generalized

Example with three Boolean features a, b, c :

Example	a	b	c	y
$\mathbf{x}_1$	1	0	1	1
$\mathbf{x}_2$	0	0	1	0
$\mathbf{x}_3$	0	1	0	1
$\mathbf{x}_4$	0	0	0	0

Goal: learn a function ϕ that predicts y given a, b, c .

Learning NNF's from Data

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Classification problem: given a dataset $\mathcal{D}$ of samples with d boolean features and one boolean target value, we want to learn an *NNF network* that approximates as closely as possible the underlying Boolean function generating these data, and predicts well on unknown data.

NNF Network - Example

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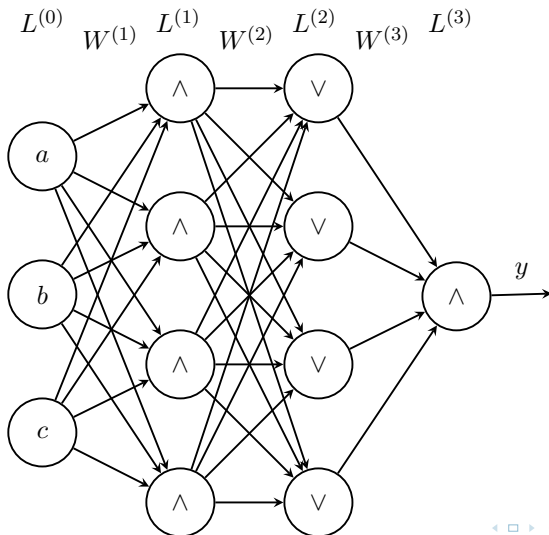
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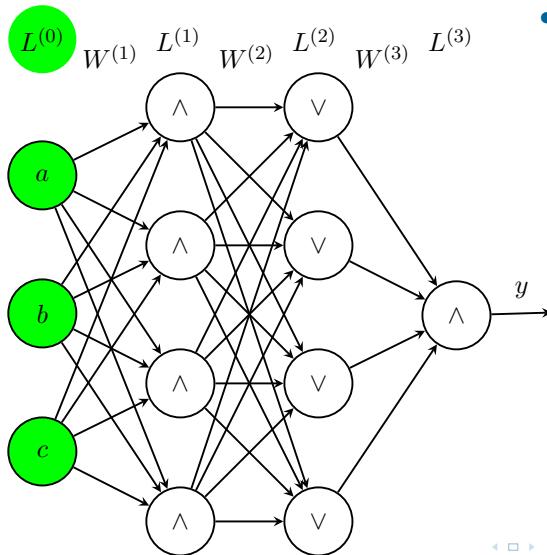
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NNF Network - Example



- $d = 3$ input features

NNF Network - Example

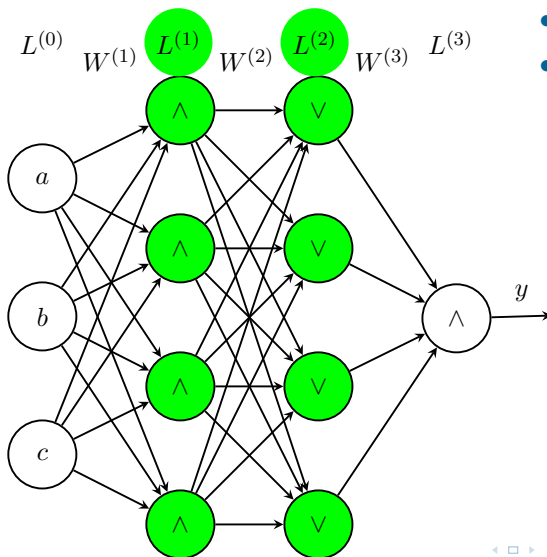
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- $d = 3$ input features
- $h = 2$ hidden layers of size $m = 4$

NNF Network - Example

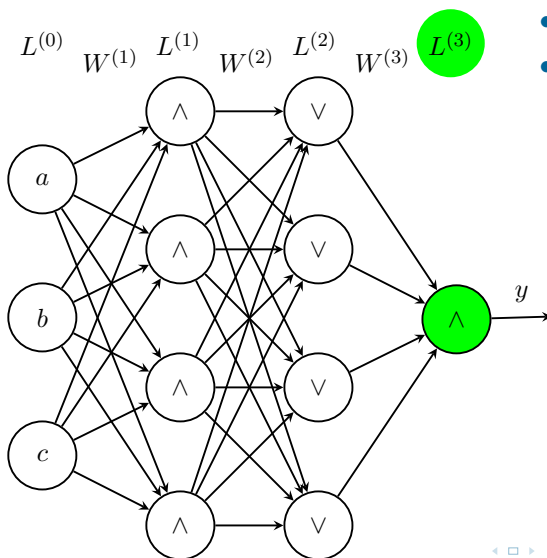
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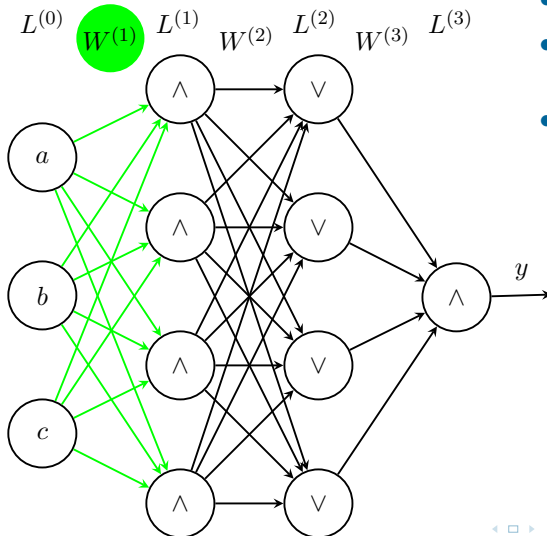
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- $d = 3$ input features
- $h = 2$ hidden layers of size $m = 4$
- $N_W = 32$ weights
 - $\dim(W^{(1)}) = 3 \times 4$

NNF Network - Example

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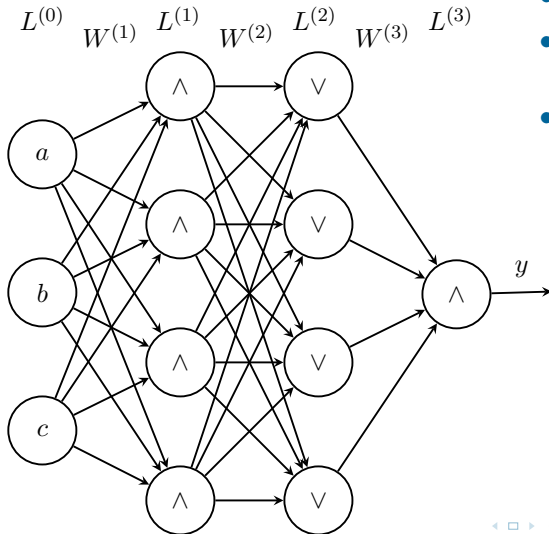
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- $d = 3$ input features
- $h = 2$ hidden layers of size $m = 4$
- $N_W = 32$ weights
 - $\dim(W^{(1)}) = 3 \times 4$
 - $\dim(W^{(2)}) = 4 \times 4$
 - $\dim(W^{(3)}) = 4 \times 1$

NNF Network - Example

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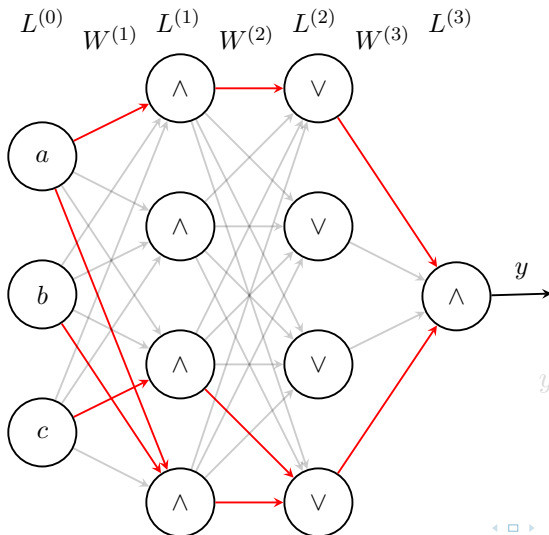
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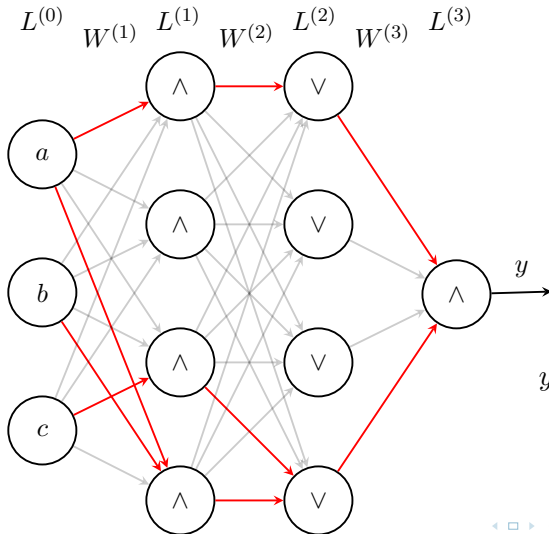
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We may view $\mathcal{D}$ as a $n \times d$ Boolean matrix, i.e., $\mathcal{D} \in \{0, 1\}^{n \times d}$.

	$\mathcal{D}$				$\mathbf{y}$
$\mathbf{x}_1$	$f_{1,1}$	$f_{1,2}$	$\dots$	$f_{1,d}$	y_1
$\mathbf{x}_2$	$f_{2,1}$	$f_{2,2}$	$\dots$	$f_{2,d}$	y_2
			$\dots$		$\dots$
$\mathbf{x}_n$	$f_{n,1}$	$f_{n,2}$	$\dots$	$f_{n,d}$	y_n

Learning an NNF with layersize $(m_\ell)_{1 \leq \ell \leq h}$ is modeled by the following SAT problem:

- Given features $f_{i,j}$ $(1 \leq i \leq n, 1 \leq j \leq d)$
- For every layer ℓ , find Boolean values
 - $w_{j,k}^{(\ell)}$ $(1 \leq j \leq m_{\ell-1}, 1 \leq k \leq m_\ell)$
- such that
 - $a_{i,k}^{(1)} = \bigwedge_{j=1}^d (w_{j,k}^{(1)} \rightarrow f_{i,j}) \wedge \bigwedge_{j=d+1}^{2d} (w_{j,k}^{(1)} \rightarrow \neg f_{i,j})$
 - $a_{i,k}^{(\ell)} = \bigwedge_{j=1}^{m_{\ell-1}} (w_{j,k}^{(\ell)} \rightarrow a_{i,j}^{(\ell-1)})$ **if** ℓ is odd $(2 \leq \ell \leq h)$
 - $a_{i,k}^{(\ell)} = \bigvee_{j=1}^{m_{\ell-1}} (w_{j,k}^{(\ell)} \wedge a_{i,j}^{(\ell-1)})$ **if** ℓ is even
 - $y_i = \bigwedge_{j=1}^{m_h} (w_j^{(h+1)} \rightarrow a_{i,j}^{(h)})$ **if** $h+1$ is odd
 - $y_i = \bigvee_{j=1}^{m_h} (w_j^{(h+1)} \wedge a_{i,j}^{(h)})$ **if** $h+1$ is even

Results - Artificial Datasets

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	NNF(5)		NNF(3)		NNF(1)		RIPPER	CART
	SAT	SHS	SAT	SHS	SAT	SHS		
Artificial Datasets								
Ø acc.	0.9857	0.9467	0.9852	0.9502	0.9939	0.9386	0.9591	0.9644
Ø rank	2.7	5.9	2.7	6.05	1.45	7	4.3	4.4
UCI Datasets								
Ø acc.	0.9682	0.9229	0.9722	0.9190	0.9807	0.9153	0.9290	0.9289
Ø rank	4.1429	4.4286	3.1429	5.7143	2.1429	5.8571	5.5714	4.2857

Figure: Average accuracies and average ranks for three different network configurations with the following layer sizes — NNF(1): [20], NNF(3): [32, 8, 2], NNF(5): [32, 16, 8, 4, 2]

Results - UCI Datasets

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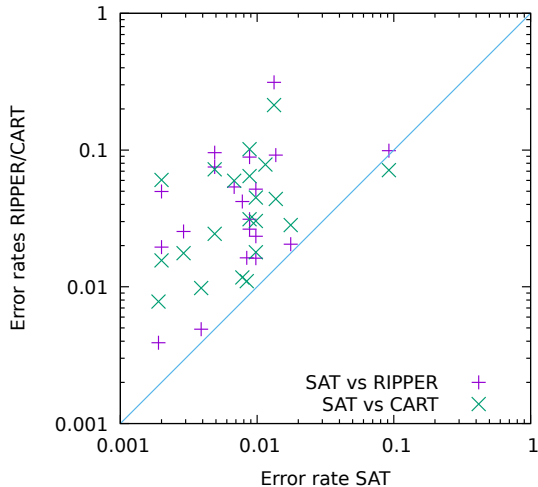


Figure: Results of SAT NNF(1) vs RIPPER and CART depicted graphically

We also used a MaxSAT solver with soft constraints $w_{j,k}^{(\ell)} = 0$ for all weights. This results in sparse networks, but is computationally expensive.

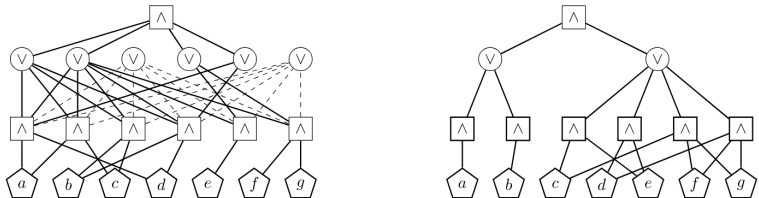


Figure: Comparison of NNF networks trained on the complete truth table for the Boolean function $(a \vee b) \wedge (c \vee d) \wedge [e \vee (f \wedge g)]$.

- Dealing with:
 - *Noise*
 - *Scalability*
 - *Overfitting*
- Learning logical networks with $\oplus = \text{XOR}$, $\otimes = \text{AND}$
- Learning probabilistic circuits
- Categorical framework for circuit learning
- Circuit complexity, finding minimal circuits